The distance covered in one rotation $= 484\ cm$

The distance covered in 35 rotations $= 484 \times 35$

$= 16940\ cm$

Example 2.6 A farmer wants to fence his circular poultry farm with barbed wire whose radius is 420 *m*. The cost of fencing is ₹12 per metre. He has ₹30,000 with him. How much more amount will be needed to fence his farm? (Here $\pi = \frac{22}{7}$)

Solution

The radius of the poultry farm is $= 420\ m$

The length of the barbed wire for fencing the poultry farm is equal to the circumference of the circle.

We know that the circumference of the circle $= 2\pi r\ units$

$= 2 \times \frac{22}{7} \times 420$

$= 2 \times 22 \times 60$

The length of the barbed wire to fence the poultry farm $= 2640\ m$

The cost of fencing the poultry farm at the rate of ₹12 per metre $= 2640 \times 12$

$= ₹31,680$

Given that he has ₹30,000 with him.

The excess amount required $= ₹31,680 - ₹30,000 = ₹1,680.$

Example 2.7 Find the perimeter of the given shape (Fig.2.9)

(Take $\pi = \frac{22}{7}$).

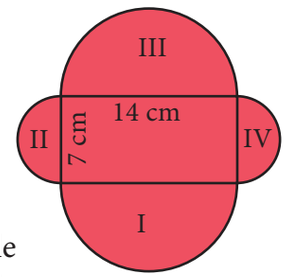


Fig. 2.9

Solution

In this shape, we have to calculate the circumference of semicircle on each side of the rectangle. The outer boundary of this figure is made up of semicircles of two different sizes. Diameters of each of the semicircles are 7 *cm* and 14 *cm*. We know that the circumference of the circle = πd *units*.

Circumference of the semicircular part $= \frac{1}{2}\pi d\ units$

Hence, the circumference of the semicircle having diameter 7 *cm* is,

$= \frac{1}{2} \times \frac{22}{7} \times 7 = 11\ cm$

Circumference of the pair of semicircular parts (II and IV) $= 2 \times 11 = 22\ cm$

Similarly, circumference of the semicircle having diameter 14 *cm* is,

$= \frac{1}{2} \times \frac{22}{7} \times 14 = 22\ cm$

Circumference of the pair of semicircular parts (I and III) $= 2 \times 22 = 44\ cm$.

Perimeter of the given shape $= 22 + 44 = 66\ cm$.

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